Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Subject: Selection Functionals Paper - My Response to Your Latest Comments

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Mon, Sep 28, 2026 at 11:35 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben,

Revision 3.1 closes the file. All three amendments are implemented.

Item 1. Taking the exact condition rather than the sufficient one was the better choice, and the proof now separates the two
statuses the old sentence ran together. The necessity is correctly scoped to $E \succeq 0$, which is where it holds.

On Corollary 3.6 being unchanged: I checked it statement by statement.
Part (i) needs $K_\Pi + \lambda K_\nu \succ 0$, and that holds for any real a because $\xi \mapsto (\xi_{m+1} - a\xi_m)_m$ is unitriangular and hence invertible.
Part (ii) uses $u > 0$, which follows from the stated hypothesis, and otherwise only $k_\bullet^2$. Your claim holds.

One thing worth adding, and it supports your choice rather than questioning it. So that "necessary" is not read further than it reaches:
the operator step fails for every $\kappa < 0$, certified by the witness at every M. The proposition's *conclusions* also fail:
a sweep over $\kappa < 0$ inside the paper's own surrogate family (positive surrogate rate,
$\sigma_\Pi = \sigma_\nu = 1$, $\lambda = 1/2$, $k_\nu \in \{0.5, 1, 2, 5\}$, k_Π from -20 to 0, $M \in \{200, 800\}$)
produces seventy-nine violations of the bounds. They do not fail across the whole half-line: along the ray $k_\nu = 2$ all three bounds hold down to
$\kappa = -19$. So there is no weaker hypothesis of the form $\kappa \geq -c$ that would save the proposition, and $\kappa \geq 0$ is the right
condition and not merely the convenient one. Nothing is asked of the manuscript; I mention it because it settles the question your amendment raises.

Items 2 and 3. Both implemented in every instance. Table 3's quadratic-variation row now carries the $O(\delta t)$, Section 10.1 drops the invalid
inference and states the L^1/L^2 distinction with the Novikov-type condition on the stochastic exponential, and the χ^2 row carries the
matching qualification.

I feel the paper is closed with no outstanding amendments.
No defect remains in the central construction, and the limits recorded in my resolution record are limits of the present results,
declared as such in the manuscript.

Ezequiel

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